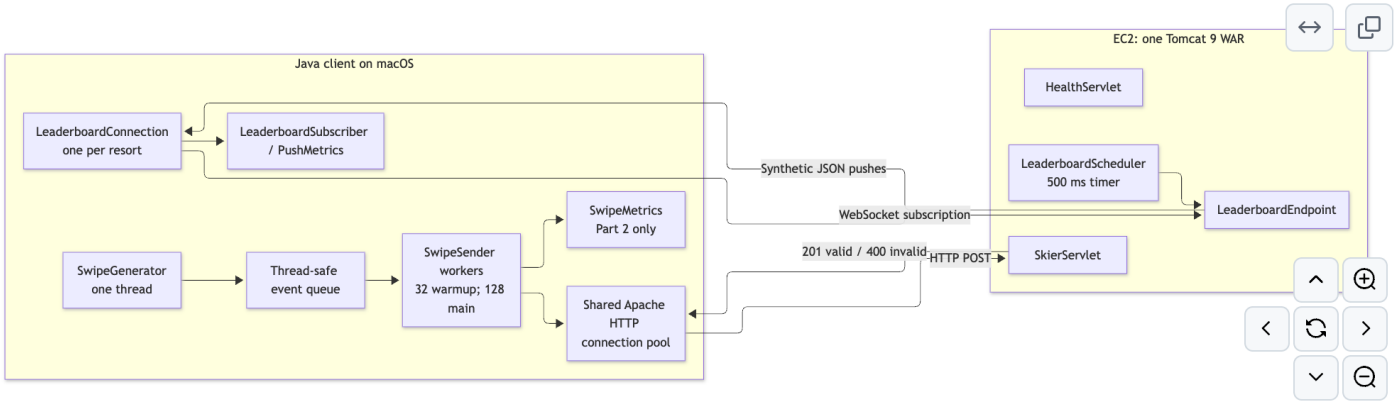
SkiTrak — Assignment 1 Design

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Architecture

The system consists of a Java load client running on macOS and one Java 11 / Tomcat 9 server deployed on EC2 in us-west-2. The server is packaged as a single WAR containing the swipe-ingest servlet, health servlet, and WebSocket leaderboard endpoint. It validates and acknowledges swipes without storing or aggregating skier data. Leaderboard messages contain synthetic standings independent of swipes.



Major classes and relationships

LoadClient coordinates event generation, subscriptions, warmup, main-phase workers, resource cleanup, and final metrics reporting.

SwipeGenerator creates immutable SwipeEvent objects and places them in an ArrayBlockingQueue.

SwipeSender removes events from the queue and sends them through the shared HttpTransport, applying application-level retries.

HttpTransport owns the Apache HTTP connection pool and records connection and request-attempt statistics.

LeaderboardConnection manages one resort subscription and schedules reconnection. Its LeaderboardSubscriber receives complete messages and records their timestamps through PushMetrics.

In Client Part 2, SwipeMetrics records HTTP attempts, exports CSV, and calculates latency statistics and throughput per resort.

On the server, SkierServlet validates swipes, HealthServlet reports status, and LeaderboardScheduler invokes LeaderboardEndpoint to broadcast synthetic standings.

Threading model

One dedicated generator thread creates all 500000 events before HTTP workers start. The main thread joins the generator and checks that the queue contains the expected number of events. Senders use nonblocking queue polling, so they never wait for event generation.

Pre-generation simplifies coordination and avoids producer starvation, at the cost of memory proportional to the number of events. Generation time is reported separately from ingest time.

Warmup uses a fixed pool of 32 workers. Each worker processes 1000 events, giving 32000 warmup events. After all warmup workers terminate, a new pool of 128 workers processes the remaining 468000. Both phases reuse the same HTTP connection pool.

Worker futures and event-count invariants detect incomplete phases. Successful and finally failed logical events are counted separately from HTTP attempts and retries.

One WebSocket subscription is established for each configured resort before ingest begins. Subscriptions remain active through both phases, and each connection has a scheduler for reconnect attempts. Cleanup closes subscriptions and the HTTP transport when the run ends.

Connection management and error handling

All HTTP senders share one Apache HttpClient connection pool. Its total and per-route capacity is max(32, mainThreads), allowing both phases to reuse connections. Responses are fully consumed before connections return to the pool.

Automatic library retries are disabled. Each logical event has one initial attempt and at most five application retries, with delays of 100, 200, 400, 800, and 1600 ms. An event is counted as failed only after all attempts fail. Retrying an ambiguous network failure can duplicate a server-side request; the assignment's dummy server does not persist events.

HTTP metrics distinguish pool occupancy, cumulative successful TCP connections, and application retries. New TCP connections opened during application retries are reported as a limited recovery metric; this does not count every replacement connection.

Each resort has a persistent WebSocket connection. Close/error callbacks schedule reconnection with exponential backoff capped at 16 seconds. Synchronized state prevents duplicate connection attempts, and generation tokens reject stale callbacks. Successful connections after the first are counted as reconnections. Shutdown cancels reconnect scheduling and aborts active sockets.

Measurement design

Client Part 2 records send and completion timestamps, monotonic elapsed time, status code, and resort ID for every HTTP attempt. It exports CSV after ingest and calculates mean, median, p95, p99, minimum, maximum, and throughput per resort. Retry delays are excluded from individual attempt latency but included in ingest runtime.

Push measurements retain the server timestamp and client receipt time for each complete, parsed message. Their difference includes clock offset between machines. Negative values are preserved and cannot be interpreted as negative network transit time.

The throughput chart counts successful responses by completion time in 10-second buckets. It includes warmup and uses the actual duration of the final partial bucket.

Little's Law and thread-count choice

The sequential Apache client measured a fresh-client request of 295.246 ms and a pooled mean response time of 27.444 ms. The fresh request includes connection and startup overhead, so it is not a pure TCP handshake measurement.

For synchronous workers with one outstanding request each, the planning estimate is $X = N / R$. With 128 workers:

$X \approx 128 / 0.027444 \approx 4664$ events/s.

The recorded main-phase results were:

Client	Main throughput (events/s)	Prediction achieved
Part 1	3644.80	78.1%
Part 2	3449.67	74.0%

Main-phase throughput is used because warmup has only 32 workers. The estimate assumes sequential response time remains representative under concurrency. Scheduling, contention, network variation, connection replacement, and client processing may increase request-cycle time. These experiments do not isolate each contributor.

128 workers was selected using earlier JDK HttpClient runs comparing 32, 64, and 128 workers. It is not claimed to be the proven optimum for Apache HttpClient.

Validation scope

Both Apache/Java 11 clients completed 500000 events with zero final failures and zero application retries. These full runs preceded the final server redeployment and minor client configuration/output changes.

Subsequent 5000-event checks verified two-resort configuration and the Part 1 connection-time summary. After server redeployment, health, selected POST boundary cases, and synthetic WebSocket pushes were verified separately. The full performance runs have not been repeated against that final WAR.

- Screenshot of Part 1 output (basic metrics, warmup reported separately)

```
--- Warmup ---
Successful: 32000
Failed: 0
Runtime: 34.639 seconds

--- Main ---
Successful: 468000
Failed: 0
Runtime: 128.402 seconds

--- Total ingest ---
Successful: 500000
Failed: 0
Remaining in queue: 0
Runtime: 163.041 seconds
Throughput: 3066.71 swipes/sec
```

```
--- HTTP transport: Apache HttpClient 5.6.4 ---
Application request attempts: 500000
Application request retries: 0
TCP connect attempts: 5059
TCP connections established: 5059
TCP connect attempts without success: 0
TCP connections established during application retries: 0
HTTP reconnection definition: new TCP opens during application retries; not all replacement/idle-expiry connections.
Pool before close: leased=0, available=128, pending=0, max=128
Sampled pool peaks (50 ms + attempt completion): leased=128, available=128, pending=0
```

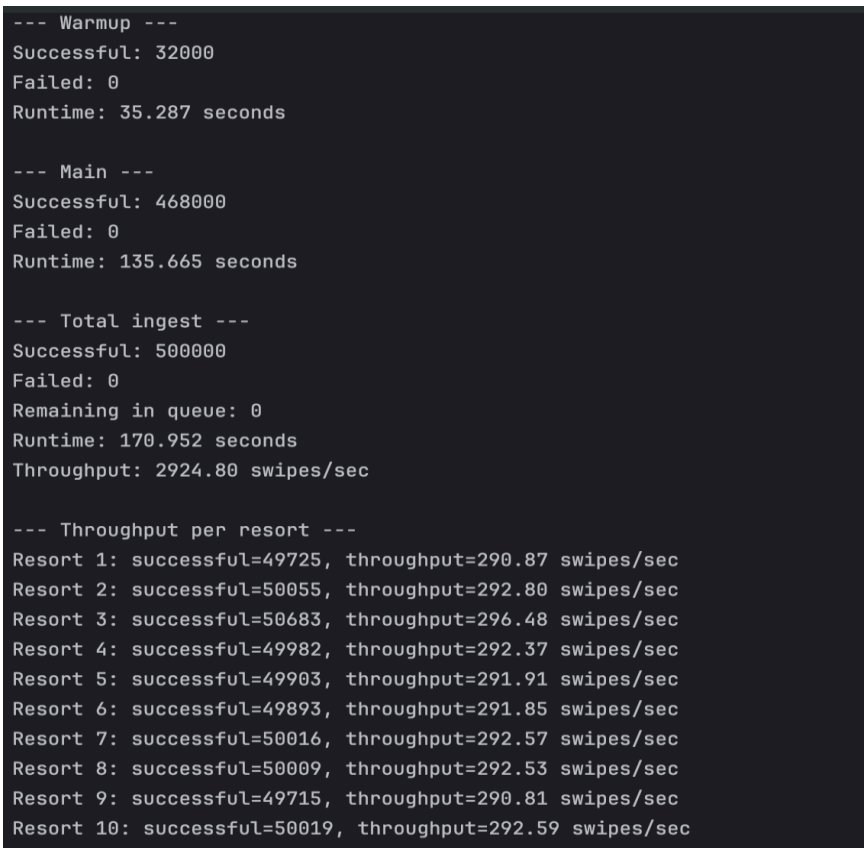
```
--- WebSocket subscriptions ---
Resort 1: updates=326, connections=1, reconnections=0
Resort 2: updates=326, connections=1, reconnections=0
Resort 3: updates=326, connections=1, reconnections=0
Resort 4: updates=326, connections=1, reconnections=0
Resort 5: updates=326, connections=1, reconnections=0
Resort 6: updates=326, connections=1, reconnections=0
Resort 7: updates=326, connections=1, reconnections=0
Resort 8: updates=326, connections=1, reconnections=0
Resort 9: updates=326, connections=1, reconnections=0
Resort 10: updates=326, connections=1, reconnections=0
Total leaderboard updates: 3260
Push CSV saved to: /Users/anne/Desktop/CS6650distributedsystem/client-part1/results/pushes-1792111343814.csv

--- WebSocket push latency ---
Samples: 3260
Mean: -72.903 ms
Median: -76.475 ms
p95: -57.196 ms
p99: 6.322 ms

Total program runtime: 165.222 seconds

Process finished with exit code 0
```

- Screenshot of Part 2 output (ingest + push-latency statistics)



```
--- Warmup ---
Successful: 32000
Failed: 0
Runtime: 35.287 seconds

--- Main ---
Successful: 468000
Failed: 0
Runtime: 135.665 seconds

--- Total ingest ---
Successful: 500000
Failed: 0
Remaining in queue: 0
Runtime: 170.952 seconds
Throughput: 2924.80 swipes/sec

--- Throughput per resort ---
Resort 1: successful=49725, throughput=290.87 swipes/sec
Resort 2: successful=50055, throughput=292.80 swipes/sec
Resort 3: successful=50683, throughput=296.48 swipes/sec
Resort 4: successful=49982, throughput=292.37 swipes/sec
Resort 5: successful=49903, throughput=291.91 swipes/sec
Resort 6: successful=49893, throughput=291.85 swipes/sec
Resort 7: successful=50016, throughput=292.57 swipes/sec
Resort 8: successful=50009, throughput=292.53 swipes/sec
Resort 9: successful=49715, throughput=290.81 swipes/sec
Resort 10: successful=50019, throughput=292.59 swipes/sec
```

```
--- HTTP transport: Apache HttpClient 5.6.4 ---
Application request attempts: 500000
Application request retries: 0
TCP connect attempts: 5058
TCP connections established: 5058
TCP connect attempts without success: 0
TCP connections established during application retries: 0
HTTP reconnection definition: new TCP opens during application retries; not all replacement/idle-expiry connections.
Pool before close: leased=0, available=128, pending=0, max=128
Sampled pool peaks (50 ms + attempt completion): leased=128, available=128, pending=0
```

```
--- WebSocket subscriptions ---
Resort 1: updates=343, connections=1, reconnections=0, avgConnectMs=771.638
Resort 2: updates=343, connections=1, reconnections=0, avgConnectMs=717.143
Resort 3: updates=343, connections=1, reconnections=0, avgConnectMs=714.754
Resort 4: updates=343, connections=1, reconnections=0, avgConnectMs=561.056
Resort 5: updates=343, connections=1, reconnections=0, avgConnectMs=702.232
Resort 6: updates=343, connections=1, reconnections=0, avgConnectMs=702.481
Resort 7: updates=343, connections=1, reconnections=0, avgConnectMs=692.326
Resort 8: updates=343, connections=1, reconnections=0, avgConnectMs=702.452
Resort 9: updates=343, connections=1, reconnections=0, avgConnectMs=701.330
Resort 10: updates=343, connections=1, reconnections=0, avgConnectMs=696.424
Total leaderboard updates: 3430
HTTP attempts recorded: 500000
CSV saved to: /Users/anne/Desktop/CS6650distributedsystem/client-part2/results/swipes-1792110301816.csv
Push CSV saved to: /Users/anne/Desktop/CS6650distributedsystem/client-part2/results/pushes-1792110301816.csv

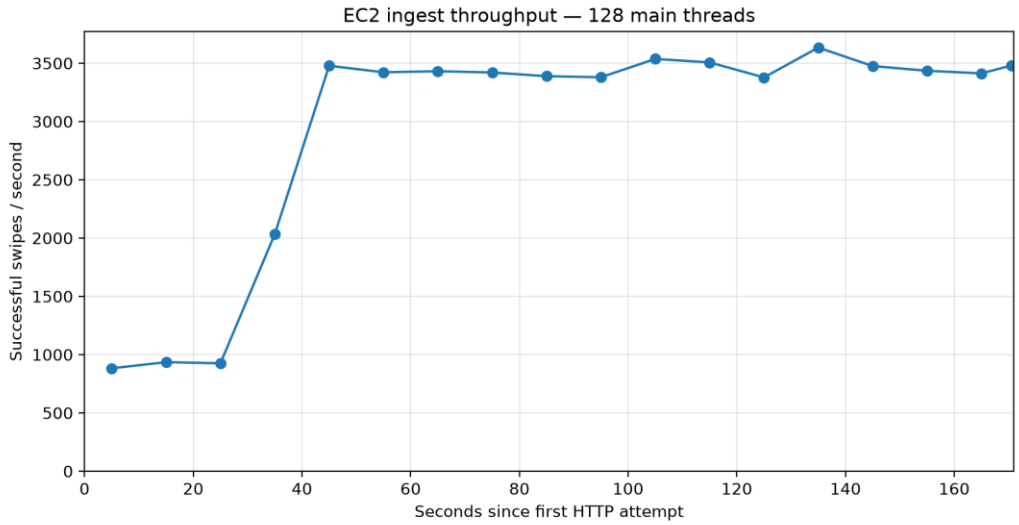
--- HTTP attempt latency ---
Samples: 500000
Mean: 36.939 ms
Median: 31.651 ms
p95: 77.010 ms
p99: 122.782 ms
Min: 11.612 ms
Max: 586.850 ms

--- WebSocket push latency ---
Samples: 3430
Mean: -90.872 ms
Median: -97.819 ms
p95: -37.446 ms
p99: -10.490 ms

Total program runtime: 173.847 seconds

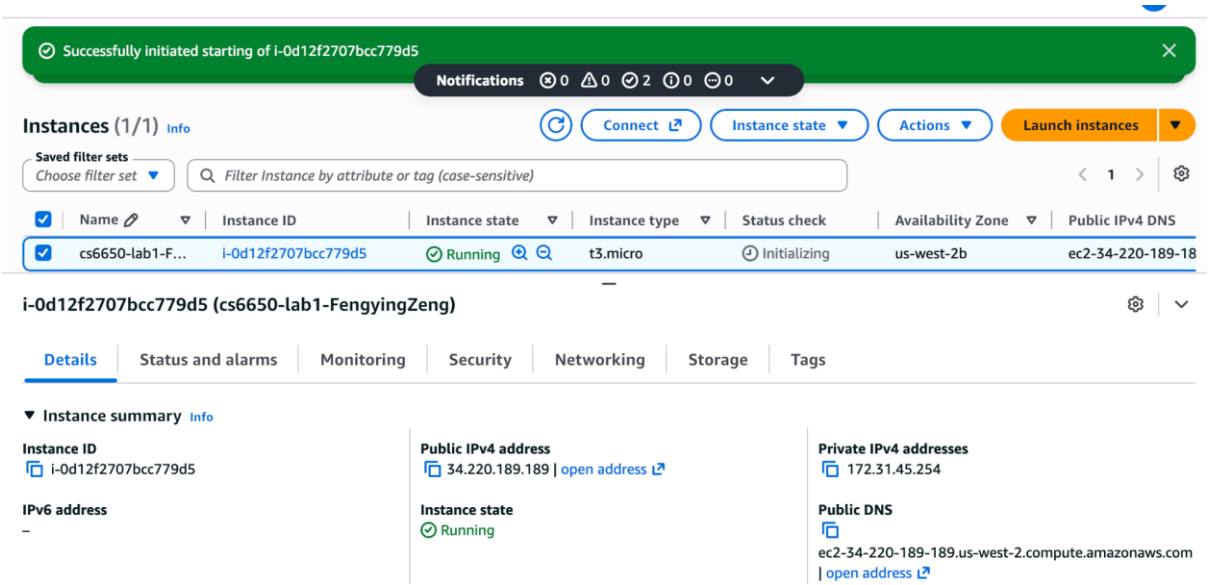
Process finished with exit code 0
```

- Throughput-over-time chart



Includes warmup. Final partial bucket uses its actual duration.

- Evidence of EC2 deployment (EC2 console screenshot)



- A screenshot of a WebSocket subscriber receiving synthetic leaderboard pushes

```
/Library/Java/JavaVirtualMachines/amazon-corretto-11.jdk/Contents/Home/bin/java -javaagent:/Applications/IntelliJ IDEA.app/Contents/lib/idea_rt.jar=55161:/Application
Receive leaderboard updates from EC2 for 10 seconds, then exit.
Connected to resort 7
Resort 7 connected in 213.658 ms
Leaderboard JSON: {"resortID":"7","serverTimestamp":"2026-10-16T03:09:34.326706Z","top":[{"rank":1,"skierID":40213,"vertical":44120},{"rank":2,"skierID":11872,"vertic
Resort 7, update 1, server time: 2026-10-16T03:09:34.326706Z, received at: 2026-10-16T03:09:34.239691Z
Leaderboard JSON: {"resortID":"7","serverTimestamp":"2026-10-16T03:09:34.826522Z","top":[{"rank":1,"skierID":40213,"vertical":44120},{"rank":2,"skierID":11872,"vertic
Resort 7, update 2, server time: 2026-10-16T03:09:34.826522Z, received at: 2026-10-16T03:09:34.799861Z
Leaderboard JSON: {"resortID":"7","serverTimestamp":"2026-10-16T03:09:35.326511Z","top":[{"rank":1,"skierID":40213,"vertical":44120},{"rank":2,"skierID":11872,"vertic
Resort 7, update 3, server time: 2026-10-16T03:09:35.326511Z, received at: 2026-10-16T03:09:35.312314Z
Updates: 19
Connections: 1
Reconnections: 0

Process finished with exit code 0
```